SUPPORTING ADULT GUIDANCE

Slide it across

Quadratics. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces the **horizontal translation** of the parabola through **$y = (x – p)^2$**. The single idea: replacing x with $(x – p)$ does not change the shape at all — it **slides the whole curve** along the x-axis. The turning point moves from (0, 0) to **(p, 0)** and the line of symmetry becomes **$x = p$**. The one genuinely counter-intuitive point — and the heart of the lesson — is the **direction**: $(x – 2)^2$ moves the curve to the **right**, because the lowest point occurs where the bracket is 0, and $x – 2 = 0$ at $x = 2$. This completes the transformation pair begun last lesson (stretch/reflect via a); combining them into $y = a(x – p)^2 + q$, and the vertex form in general, is later work in the pathway.

Driving Question: *Where does the curve's lowest point go, and why?*

The mathematics, in plain terms

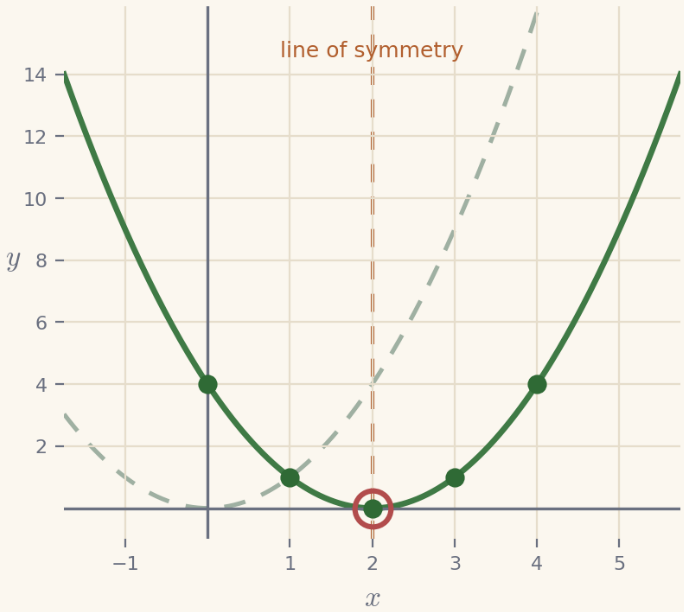
$y = (x – p)^2$ is the ordinary parabola $y = x^2$ **slid sideways**. Nothing about its shape changes — it is not taller, wider or flipped — it just sits in a different place left-to-right. The learner can see this from a table: the heights are identical to $y = x^2$, simply lined up under different x-values. The only thing to get straight is the **direction**, which surprises almost everyone at first: **$(x – 2)^2$ moves to the right**, not the left. The reliable way to see why is to ask *where is the bottom of the curve?* The bottom is where the squared bracket is smallest — that is, 0 — and $(x – 2)$ is 0 when $x = 2$. So the lowest point sits at **(2, 0)**, and the whole curve has shifted 2 to the right. From there: the **turning point** is **(p, 0)** and the **line of symmetry** is the vertical line **$x = p$**. The most useful support is to keep returning to "set the bracket to zero" whenever the direction feels wrong — it turns a confusing rule into a reliable check.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — slide it right — y = $(x - 2)^2$

Make a table for y = $(x - 2)^2$ by working out the bracket first, then squaring. The dashed curve is y = x^2 for comparison.

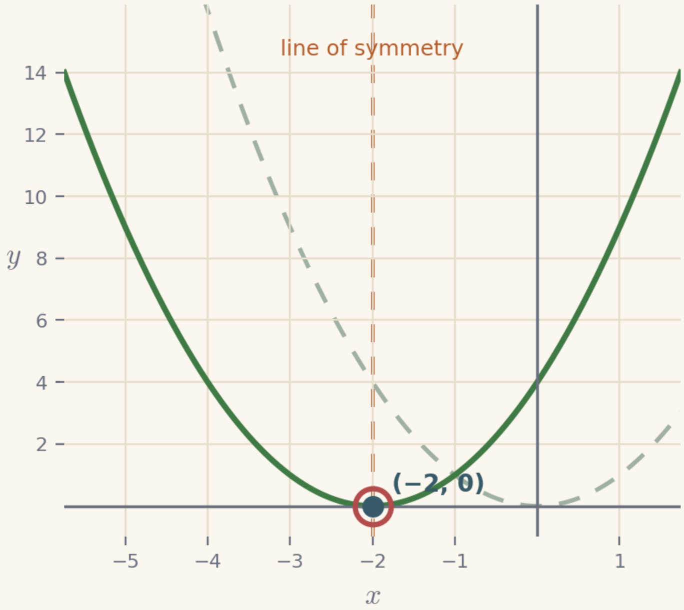


1. $x = 0$	$(0 - 2)^2 = (-2)^2 = 4$
2. $x = 2$	$(2 - 2)^2 = 0^2 = 0$
3. $x = 4$	$(4 - 2)^2 = 2^2 = 4$
4. lowest point	where the bracket is 0: $x = 2$

the curve is y = x^2 slid 2 to the right; turning point (2, 0), line of symmetry $x = 2$

Example 2 — write the equation for a lowest point

Find the rule y = $(x – p)^2$ whose lowest point is at (–2, 0). The bracket must be 0 there.



1. need the bracket 0 at $x = –2$
2. so the bracket is $x – (–2) = x + 2$
3. the rule is $y = (x + 2)^2$
4. check at $x = –2$ $(–2 + 2)^2 = 0$

y = $(x + 2)^2$, with turning point (–2, 0) and line of symmetry $x = –2$

Set the bracket to zero to find the bottom

Two moves carry this lesson. First, **slide by a table**: work out $(x – p)^2$ for a few x-values and notice the heights are just y = x^2 relocated — the shape is untouched, only the position. Second, and most important, **set the bracket to zero**: the lowest point is where $(x – p) = 0$, i.e. $x = p$, so the turning point is (p, 0) and the line of symmetry is $x = p$. Make this the learner's reflex whenever the left/right direction feels backwards — it is the reason underneath the surprise, and it never lets them down. Keep the sign explicit: $(x + 1)^2$ is $(x – (–1))^2$, so its bottom is at $x = –1$. Resist combining this with the stretch a from last lesson for now; the two transformations are joined later. The stretch here — writing the equation for a chosen turning point — is the same idea run backwards: to place the bottom at x = 3, make the bracket 0 there, so subtract 3.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	The same curve, moved sideways: a table for y = $(x - 2)^2$, plotted beside y = x^2.	~9 min
Movement	The p-slider: move the dial and watch the curve, turning point and line of symmetry slide together.	~14 min
Reflection	Find the bottom: read the turning point (p, 0) from an equation; mind the sign.	~10 min
Creativity	Write the equation (stretch): choose p so the lowest point lands where you want.	~8 min
Rest	A pause after moving a whole curve with one number.	~2 min
Nutrition	Mode B (intellectual): what the learning feeds — trusting structure (bracket = 0) over a first-impression hunch.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking $(x - 2)^2$ slides the curve to the left

Repair: "Ask where the bottom is. The bracket is 0 when $x - 2 = 0$, i.e. $x = 2$ — a positive x — so the curve moves right to (2, 0)."

Reading the turning point of $(x + 1)^2$ as $x = 1$

Repair: "$(x + 1)$ is $(x - (-1))$. The bracket is 0 when $x = -1$, so the turning point is (–1, 0)."

Believing the curve changes shape as well as position

Repair: "Check the heights in the table — they are identical to $y = x^2$. Only the position moves; the shape is the same."

Thinking the line of symmetry is horizontal

Repair: "It is vertical and passes through the turning point: for $(x - p)^2$ it is the line $x = p$."

Squaring the bracket wrongly, e.g. $(x - 2)^2 = x^2 - 4$

Repair: "$(x - 2)^2$ means $(x - 2)(x - 2)$. For now, just substitute a value: at $x = 0$ it is $(-2)^2 = 4$, not –4."

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: plotting $y = x^2$ (Lesson 5), substituting into a bracket, and squaring including squaring a negative. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner sliding the curve and setting the bracket to zero:

- "Work out $(x – 2)^2$ at $x = 0, 1, 2, 3, 4$. Where is the lowest point?"
- "Which way does $(x – 5)^2$ move the curve, and how far?"
- "$(x + 3)^2$ — where is its turning point? Watch the sign."
- "Stretch: what is p if the lowest point is at (4, 0)? What about (–1, 0)?"

If they stall, that is the moment for a Reconnection Route or the lesson's prompt and hint — not for the answer.

Questions you might have

"Why does $(x – 2)^2$ move the curve right, not left?"
Because the lowest point is where the bracket is 0, and $x – 2 = 0$ at $x = 2$. So the bottom of the curve sits at $x = 2$ — it has moved 2 to the right. Setting the bracket to zero is the reliable way to see the direction.

"What are the turning point and line of symmetry of $y = (x – p)^2$?"
The turning point is (p, 0) and the line of symmetry is the vertical line $x = p$ — both found by making the bracket 0.

"Do they combine this with the stretch a from last lesson?"
Not yet. Keeping $(x – p)^2$ on its own lets the slide be understood cleanly; $y = a(x – p)^2$ and the general vertex form come later in the pathway.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.

Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.